

Exercises on Triggers

Consider the following database for a bank that maintains Customers, Accounts, and Branches.

Customers: CustomerID, Name, City

Branches: BranchID, BranchName, Location

Accounts: AccountID, CustomerID, BranchID, Balance

Customers		
CustomerID	Name	City
1	Alice	Colombo
2	Bob	Kandy
3	Charlie	Galle
4	David	Jaffna
5	Eva	Negombo
6	Frank	Kurunegala
7	Grace	Matara
8	Hannah	Anuradhapura
9	Ishan	Batticaloa
10	Julia	Trincomalee
11	Kevin	Ratnapura
12	Lara	Badulla

Branches		
BranchID	BranchName	Location
1	Main Branch	Colombo
2	City Branch	Kandy
3	Coastal Branch	Galle
4	North Branch	Jaffna
5	Airport Branch	Negombo
6	Central Branch	Kurunegala
7	South Branch	Matara
8	Ancient Branch	Anuradhapura
9	East Branch	Batticaloa
10	Harbor Branch	Trincomalee

11	Gem Branch	Ratnapura
12	Hill Branch	Badulla

Accounts			
AccountID	CustomerID	BranchID	Balance
101	1	1	50000
102	2	2	30000
103	3	3	45000
104	4	4	20000
105	5	5	35000
106	6	6	60000
107	7	7	15000
108	8	8	25000
109	9	9	40000
110	10	10	55000
111	11	11	70000
112	12	12	80000
113	1	2	22000
114	3	1	33000
115	5	3	27000

Consider the Bank database created for the previous exercise (Exercise on Transactions) and answer the following questions.

1. Write a trigger to prevent a negative balance before inserting a new record into the Account relation.
2. Write a trigger to set the default balance if it is null before inserting a new record into the Account relation.
3. Write a trigger to ensure the customerID exists before inserting a new record into the Account relation.
4. Write a trigger to ensure the banchID exists before inserting a new record into the Account relation.
5. Write a trigger to log new account creation after inserting a new record into the Account relation.
6. Write a trigger to log new customer after inserting a new record into the Customer relation.

7. Write a trigger to alert account creation with more than 100,000 after inserting a new record into the Account relation.
8. Write a trigger to prevent customerID from changing before updating a record in the Customer relation.
9. Write a trigger to ensure the branch exists before updating a record in the Account relation.
10. Write a trigger to limit the balance increase to not more than 10 times before updating a record in the Account relation.
11. Write a trigger to log the account balance update after updating a record in the Account relation.
12. Write a trigger to log the customer update after updating a record in the Customer relation.
13. Write a trigger to alert to large withdrawals (above 100,000) after updating a record in the Account relation.
14. Write a trigger to log the branchID change after updating a record in the Account relation.
15. Write a trigger to prevent deleting active accounts before delete from the Account relation.
16. Write a trigger to prevent deleting customers with accounts before delete from the Customer relation.
17. Write a trigger to create a backup entry before deleting from the Account relation.
18. Write a trigger to prevent deleting branches with accounts before deleting from the Customer relation.
19. Write a trigger to log delete attempts before deleting from the branch relation.
20. Write a trigger to log account deletion after deletion from the Account relation.
21. Write a trigger to log customer deletion after deletion from the Customer relation.
22. Write a trigger to log branch deletion after deletion from the Branch relation.
23. Write a trigger to alert high-value (above 100,000) account deletion after deletion from the Account relation.